AMIS 1012 Ethics in Computing

Chapter 5: Impact of IT on Quality of Life

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Learning Objectives

1. To discuss recent technological advances on quality of life
2. To study the advancements in health care systems

Chapter Outline

1. Information and Communication Technology
2. Telemedicine and Telehealth
3. Mobility
4. Electronic Health Records
5. Big Data and Cloud Computing
6. Privacy in Digital Technology
7. Internet and the Web
8. Online Identity.
9. Facial Recognition and Privacy Rights
10. Fingerprint and Applied Ethics
11. Smartphone Applications
12. Social Media
13. Major laws on Privacy

Introduction

- IT has played an important role in encouraging creativity and is a crucial factor in growing efficiency.
- IT, as well as other innovative technologies and financial spending, was used by progressive management teams to introduce product, operation, and service advancements.
- Productivity increases were simple to calculate. Example: Company replaced a number of accountants when electronic payroll introduced.

1.Information and Communication Technology (ICT)

- **ICT** is the **integration** of telephone and computer networking in a single cabling device that makes for **simple data collection, exploitation, control, and retrieval**.

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1.Information and Communication Technology (ICT)

Importance of ICT

- **Strengthens students' critical thinking**, helping them to research and discover proper solutions to challenges in all relevant fields.
- **Encourages students to be creative** and create innovative answers.
- Enables th**e storing and retrieval of data**.
- **Improves computer networking**, which is now referred to as the Internet and intranet.

1.Information and Communication Technology (ICT)

Importance of ICT

- **Boosts national economic development** since it is a reliable source of revenue for all nations that recognize its significance.
- Creates lucrative **employment** and raises awareness of other issues, making it a viable source of income.
- Provides a forum for local and worldwide academics in the field of IT to **exchange ideas and developments**.

1.Information and Communication Technology (ICT)

Importance of ICT

- Serves as a **platform for e-learning** as well as an online library. (Distributing information is easier)
- Essential to **globalization**.
- Used in various departments to **keep records** of government operations and administration.

2. Telemedicine and Telehealth

- Instead of personally attending a doctor’s office or hospital, **telemedicine helps patients to connect with a healthcare provider** through technology.

- Using video, web portals, and email, patient can discuss conditions and medical problems in **real time** from a healthcare professional.

2. Telemedicine and Telehealth

Get diagnosis

Get information on medical choices

Get prescription

2. Telemedicine and Telehealth

- Telemedicine can be divided into **three categories**

Category	Description
Immersive medicine	Also known as “ live telemedicine”. A form of telemedicine in which **doctors and patients collaborate in real time**.
Remote patient monitoring	Helps nurses to keep an eye on patients and **use handheld medical devices to record data** such as blood pressure and blood sugar levels.
Store and forward	Telemedicine providers may **share a patient’s confidential records with other physicians** or **experts**.

2. Telemedicine and Telehealth

- Telemedicine has expanded to deliver treatment in a number of areas as technology has progressed.
- Nowadays, personal physician can run websites, use video conference technology for remote appointments. Telemedicine providers will manage the applications.
- Suitable only for **non-emergency situations**. (Eg: minor problems and follow-up appointments)

2. Telemedicine and Telehealth

- Examples of telemedicine usage:
  - arrange a video meeting with your healthcare provider to address your symptoms if you think a cut is contaminated.
  - If you're on break and believe you're having strep throat, you should call your primary care provider.
  - It may be used for a number of other health needs, such as psychotherapy and tele dermatology, which includes consultations on moles, rashes, and other skin disorders.
  - Other general conditions treated by telemedicine include colds and flu, bug bites, sore throats, diarrhea, and pink eye.

2. Telemedicine and Telehealth

- Telemedicine is **not appropriate for emergency situations** involving X-rays, splints, or casts, such as a heart attack or stroke, wounds or lacerations, or broken bones requiring X-rays, splints, or casts.
- Every condition that necessitates urgent, hands-on attention should be dealt with in person.

2. Telemedicine and Telehealth

Telemedicine vs. telehealth: What's the difference

Telemedicine	Telehealth
Patient **get medication without having to have an appointment with a doctor** or visit their clinic for emergency care.	It's a method of bettering medical safety and specialist education. Telehealth encompasses **nonclinical events** such as appointment preparation, continued medical school, and specialist training, in addition to telemedicine.

2. Telemedicine and Telehealth

Advantages of Telemedicine

Advantages	Description
Comfort and convenience	Patient **don't have to drive to the clinic**. Virtual visits can be easier to arrange especially for those who have tight schedule.
Control of Infectious Illness	Doctors can **prescreen patients** for possible infectious disease. Less exposure of the disease to other people.
Better Assessment	Doctors can **see patients in their home environment**. For example, identify source of allergies and mental health assessment.
Family Connections	During consultations, a **family member can help to provide information**, ask questions and take note of the doctor's answers.

2. Telemedicine and Telehealth

- Telemedicine has shown to be more beneficial in the area of mental health than in any other medical field. Those in need of mental assistance will now connect with a psychiatrist at the touch of a button, for a fraction of the cost of a full office appointment.

3. Mobility

- Mobility is changing healthcare today. Smart devices and applications are not only empowering health agencies, but they are also greatly changing the patient/ provider relationship.
- **Mobility and cloud access** have significantly assisted in expanding connectivity for both patients and physicians. Instead of using paper charts, hospitals, insurance providers, and physicians' offices are also **keeping patient care data in the cloud**, with patients **able to view examination results online** 24 hours a day, 7 days a week.

3. Mobility

- Mobility has been a great enabler for Telemedicine, whether it’s live streaming of medical data from a connected computer, high-fidelity audio communication over a handheld device, or information sharing from an Internet of Things (IoT) medical device.

3. Mobility

- Providers can offer better treatment due to mobility.

Advantages of mobility
Improving efficiencies
Improving the patient experience both within and outside the hospital
Improving contact and coordination between patients and healthcare professionals
Patient tracking using mobile devices is expected to save substantial healthcare costs in the immediate future.

3. Mobility

- The enormous simplicity of recording, exchanging, reviewing, and collaborating medical knowledge, the widespread use of mobility technologies is **promoting faster implementation of telemedicine applications**.
- An increasing variety of medical apps for mobile devices is assisting in the development of telemedicine for both doctors and patients. As a result, **hospitals are becoming more effective, and patient conditions are improving**.

4. Electronic Health Records

- Electronic Health Records (EHR) is a computerized representation of a patient's paper medical chart. EHRs are **real-time, patient-centered records** that make information readily and securely accessible to authorized users.
- An EHR framework may give a **patient's medical and prescription history**, it is intended to go beyond the traditional health reports gathered in a provider's office to provide a more complete picture of a patient's care.

4. Electronic Health Records

- **Features** of an EHR:
  - archive a patient’s medical history, diagnosis, prescriptions, recovery schedules, vaccination dates, allergies, radiology images, and diagnostic and test results.
  - provide patients access to evidence-based resources for making clinical decisions.
  - simplify and streamline physician workflow.

4. Electronic Health Records

- **Benefits of EHR**

Benefits	Description
Better quality of care	• EHR enables doctors to provide excellent service to their patients by making patient information easily accessible, resulting in more effective treatment. • EHRs include health analytics that assist physicians in identifying patterns, forecasting diagnoses, and recommending treatment choices. • Patients may visit patient websites to view previous medical data such as lab and imaging results, medications, diagnoses, and other information. Patients may communicate with their physicians by sharing information, sending text messages, or even participating through video conferencing.

4. Electronic Health Records

- **Benefits of EHR**

Benefits	Description
Willingness to communicate with one another	Combining EHR with other systems such as Electronic Medical Record (EMR) system, may help hospitals improve treatment quality. When patients continue to visit physicians, manage chronic illnesses such as diabetes, or plan to move to a home health environment, an interoperable EHR system is critical.
Increased productivity	EHR allows doctors to offer more precise care and diagnosis. It reduce wait times for appointments and office visits by maintaining a patient-centered approach, allowing health providers to treat more patients on a regular basis.

4. Electronic Health Records

- EHRs enable seamless interchange of data through a contemporary healthcare system that embraces and utilizes digital innovation and has the potential to alter how treatment is delivered and reimbursed.

- Knowledge can be obtained anytime and wherever it is required with EHRs.
    - **Increased patient participation**
    - **Improved care coordination**
    - **Improved diagnostics and patient outcomes**
    - **Practice efficiencies and cost savings**

5. Big Data and Cloud Computing

- In a number of sectors, big data has become a catchphrase. This is due to the **ability to generate and collect massive amounts of data from a variety of sources.**
- The collected data is utilized in **analytics**, which **allows for future prediction** (**Eg:** the early identification of future disease outbreaks and, ultimately, the prevention of fatalities.)

5. Big Data and Cloud Computing

- Data management in the cloud improves productivity and accessibility thereby minimizing waste.
- From the healthcare perspective, cloud computing aids research and development in the creation of new treatment regimens and life-saving drug compositions.
- Cloud services, in fact, may be very helpful to medical care, **allowing for enormous amounts of research and study, as well as efficient clinical data exchange.**

5. Big Data and Cloud Computing

- Without the hassle or price of maintaining additional server hardware, the cloud provides **dependable and cost-effective storage choices**, as well as backup and recovery capabilities.
- The healthcare business can upload more data to the cloud, and Big Data analytics can extract insights from that data, paving the way for a more progressive future for the industry.

5. Big Data and Cloud Computing

Introduction to Cloud Computing

- For large-scale storage and sophisticated analysis, as demanded by Big Data and Big Data Analytics, cloud computing is the most suitable architecture as it offers flexibility, security, parallel processing, scalability, and resource virtualization.
- Cloud storage can **lower** automation, information, and infrastructure maintenance **costs** while also **improving operating performance and user access.**

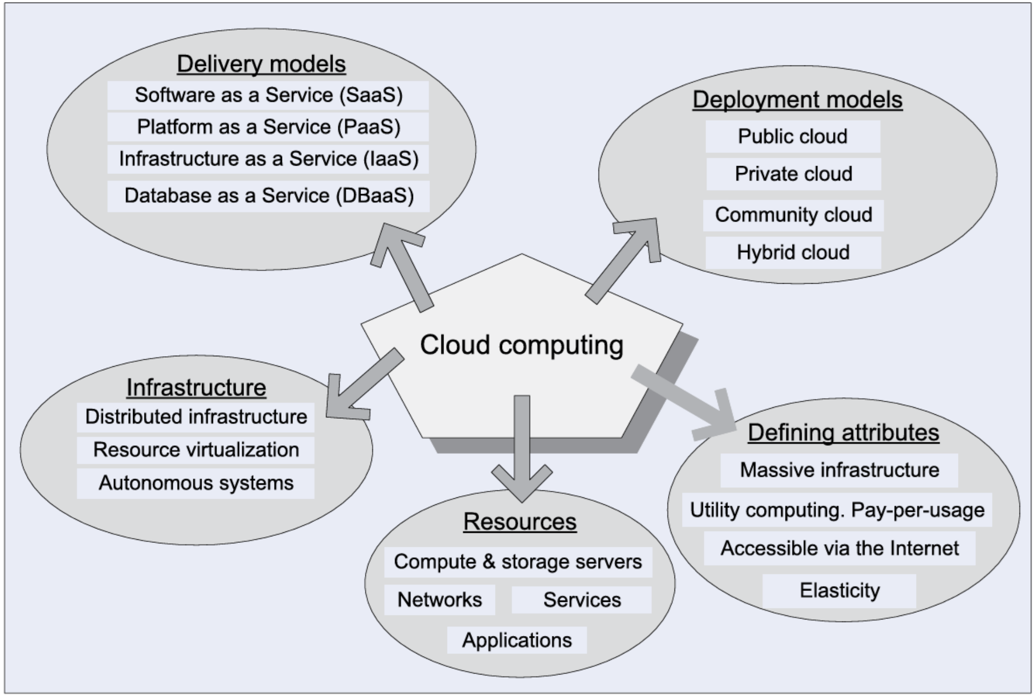
5. Big Data and Cloud Computing

Introduction to Cloud Computing

- Cloud computing characteristics:

Characteristics	Description
Uses Internet technologies to offer elastic services	The ability of **dynamically acquiring computing resources** and supporting a variable workload. A cloud service provider maintains a massive infrastructure to support elastic services.
Resources used for these services can be metered.	Users **charged only for the resources they used**.
User convenience	**Maintenance and security are ensured by service providers.**

5. Big Data and Cloud Computing



A summary view of cloud computing: delivery models, deployment models, defining attributes, and organization of cloud infrastructure.

5. Big Data and Cloud Computing

Introduction to Cloud Computing

- How cloud computing is affecting healthcare:
  - **Cost-cutting**
    - Cloud computing enables healthcare providers to access data storage and processing **without upfront costs for hardware**.
    - This model is cost-effective and scalable, efficiently handling large volumes of patient data from various sources while minimizing expenses.

5. Big Data and Cloud Computing

Introduction to Cloud Computing

- How cloud computing is affecting healthcare:
  - **Interoperability ease**
    - Cloud-based interoperability enhances healthcare by **linking data from various sources**, making patient information readily accessible.
    - This improves planning, execution, and collaboration among providers. It also streamlines data sharing among pharmaceuticals, insurers, and payment systems, speeding up healthcare delivery and improving quality.

5. Big Data and Cloud Computing

Introduction to Cloud Computing

- How cloud computing is affecting healthcare:
  - **Links to specialized analytical methods**
    - Cloud computing allows for the aggregation and analysis of both structured and unstructured healthcare data.
    - Leveraging Big Data Analytics and AI, cloud technology **enhances medical research and enables the development of personalized treatment plans**. It ensures comprehensive documentation of medical information and facilitates effective data mining for specific health insights.

5. Big Data and Cloud Computing

Introduction to Cloud Computing

- How cloud computing is affecting healthcare:
  - **Data possession by the patient**
    - Cloud computing empowers individuals to make informed health decisions by decentralizing data and **enhancing patient engagement**.
    - It allows for quick storage and retrieval of patient data and medical images, with improved reliability and data recovery due to automatic backups and reduced data redundancy, despite ongoing security concerns.

5. Big Data and Cloud Computing

Introduction to Cloud Computing

- How cloud computing is affecting healthcare:
  - **Skills of telemedicine**
    - Cloud computing enhances healthcare by enabling remote data access, which improves telemedicine, post-hospitalization care, and virtual drug adherence.
    - It **increases healthcare accessibility and patient experience** through seamless data sharing, better connectivity, and comprehensive coverage during prevention, treatment, and rehabilitation.

6. Privacy in Digital Technology

- A cell phone is a marvel that provides a camera, microphone, and minuscule sensors ready to share your life with the world. With these new tools come new challenges to old ways of thinking.
- What does it mean to preserve privacy in a social media world where every experience is recorded and preserved? Who owns the memories we create with these devices? Who has the right to share them? To delete them? To profit from them?

6. Privacy in Digital Technology

- Privacy concerns the **collection and use of data about individuals**. There are three primary privacy issues:
  - **Accuracy** relates to the responsibility of those who collect data to **ensure that the data is correct.**
  - **Property** relates to **who owns data**.
  - **Access** relates to the responsibility of those who have data to control **who is able to use that data**.

6. Privacy in Digital Technology

- Almost all human events are recorded digitally.
  - For example, making a phone call, posting a video to social media, or wearing a smartwatch creates a digital record. A digital record of a phone call can include who you called, when you called, where you were when you made the call, and even the contents of the conversation itself.
  - This results in an unprecedented amount of digital information being stored. This stored information is often referred to as **big data.**

6. Privacy in Digital Technology

- Large organizations are constantly compiling information about us. The federal government alone has over 2,000 databases.
  - For example, credit card companies maintain user databases that track cardholder purchases, payments, and credit records.
  - Supermarket scanners in grocery checkout counters record what we buy, when we buy it, how much we buy, and the price.
  - Financial institutions, including banks and credit unions, record how much money we have, what we use it for, and how much we owe.
  - Search engines record the search histories of their users, including search topics and sites visited. Social networking sites collect every entry.

6. Privacy in Digital Technology

- This collection of data can be searched to find all the action of one person—creating a **digital footprint** that reveals a highly detailed account of your life. A vast industry of data gatherers known as **information resellers or information brokers** now exists that collects, analyzes, and sells such personal data.

6. Privacy in Digital Technology

- Digital footprint can reveal more than you might wish to make public and have an impact beyond what you might imagine. This raises many important issues, including:

    - **Collecting public, but personally identifying, information**
    - **Spreading information without personal consent**
    - **Spreading inaccurate information**

7. Internet and the Web

- Every computer on the Internet is identified by a unique number known as an **IP address**.
- IP addresses can be used to **trace Internet activities** to their origin, allowing computer security experts and law enforcement officers to investigate computer crimes such as unauthorized access to networks or sharing copyright files without permission.

7. Internet and the Web

- Some websites are designed to be hidden from standard search engines. These websites make up the deep web and allow communication in a secure and anonymous manner.
- One part of the deep web is hidden websites that make up the **dark web**. These websites use special software that hides a user's IP address and makes it nearly impossible to identify who is using the site.
- The ability to communicate anonymously attracts criminals who want to sell drugs, or profit from the poaching of endangered animals.

7. Internet and the Web

- When we browse the web, the browser stores critical information, without us being aware of it. This information, which contains records about our Internet activities, includes history and temporary Internet files.
  - **History files** include the locations, or addresses, of sites that recently visited. This history file can be displayed by browser in various locations, including the address bar and the History page.
  - **Temporary Internet files** (browser cache), contain web page content and instructions for displaying this content. Whenever we visit a website, these files are saved by browser. If we leave a site and then return later, these files are used to quickly redisplay web content.

7. Internet and the Web

- **Cookies** - method to monitor web activity. It is a small data files that deposited on hard disk from visited websites. Based on the browser's settings, these cookies can be accepted or blocked.

- **Benefit** - giving personalized experiences on the web. Although cookies are harmless, what makes them a potential privacy risk is that they can store information about you, your preferences, and your browsing habits. The information stored generally depends on whether the cookie is a **first-party** or a **third-party cookie**.

7. Internet and the Web

- **First-party cookie** - generated (and then read) only by the website you are currently visiting.
  - Many websites use first-party cookies to store information about the current session, your general preferences, and your activity on the site. The intention of these cookies is to provide a personalized experience on a particular site.
  - For example, when you are shopping online, you might fill your cart with items and then leave the site without making making a purchase. Cookies allow you to return to that site and have the site remember what was in your cart.

7. Internet and the Web

- A **third-party cookie** is usually generated by an advertising company that is affiliated with the website you are currently visiting.
  - These cookies are used by the advertising company to keep track of your web activity as you move from one site to the next. For this reason, they are often referred to as tracking cookies.
  - For example, suppose you visit four different websites that employ the same advertising agency. The first three sites are about cars, but the fourth is a search engine. When you visit the fourth site, you will likely see a car advertisement because your cookie showed that you had been visiting car-related websites.

7. Internet and the Web

- Some users are not comfortable with the idea of web browsers storing so much information in the form of temporary Internet files, cookies, and history.
- For this reason, browsers now offer users an easy way to delete their browsing history.
- In addition, most browsers also offer a privacy mode, which ensures that your browsing activity is not recorded on your hard disk. For example, Google Chrome provides **Incognito Mode** accessible from the Chrome menu, and Safari provides **Private Browsing** accessible from the Safari option on the main menu.

8. Online Identity

- Another aspect of Internet privacy comes from **online identity**, the information that people **voluntarily post** about themselves online.
- With the popularity of social networking, blogging, and photo- and video-sharing sites, many people post intimate details of their lives without considering the consequences.
- Although it is easy to think of online identity as something shared between friends, the archiving and search features of the web make it available indefinitely to anyone who cares to look.

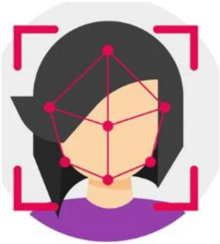
8. Online Identity

- There are any number of cases of people who have lost their jobs on the basis of posts on social networking sites.
- These job losses range from a teacher (using offensive language and photos showing drinking) to a chief financial officer of a major corporation (discussing corporate dealings and financial data). The cases include college graduates being refused a job because of Facebook posts.
- How would you feel if information you posted about yourself on the web kept you from getting a job?

9. Facial Recognitions and Privacy Rights

9.1 Facial Recognition

- Contemporary biometric facial recognition is a digitalized extension of facial mapping, utilising an algorithm to undertake the comparison.
- In a similar process to that described in fingerprint identification, it is a digital comparison of the arrangement of facial features.



9. Facial Recognitions and Privacy Rights

- The process started with a digital photograph being taken **(capture)**, and the face scaled and aligned to establish a baseline position **(process)**.
- The facial features are then quantified to create a contour map of the position of individual facial features that is converted into a digital template **(modelling)**.

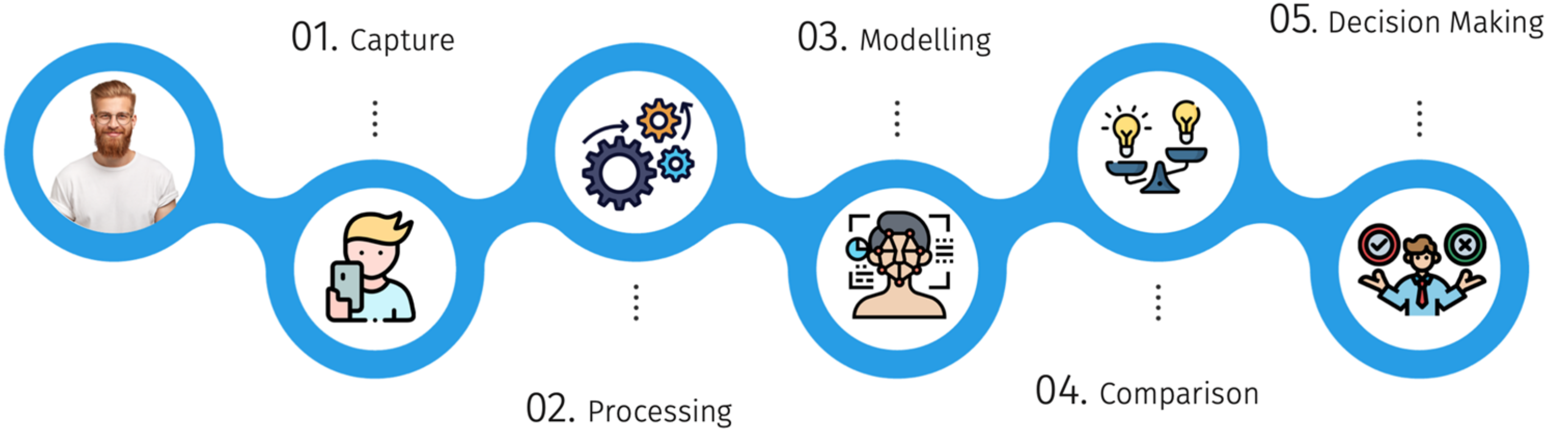
9. Facial Recognitions and Privacy Rights

- In the matching process, pairs of digital templates are compared **(comparison)**, and a numerical score derived, representing a probabilistic measure that they are of the same person **(decision making).**

9. Facial Recognitions and Privacy Rights

- The process of verification is undertaken through one-to-one matching: the live comparison of a face with a digital template stored in an identity document, such as a person presenting a passport at border control.

9. Facial Recognitions and Privacy Rights



The process of facial recognition

https://www.mobbeel.com/en/the-ultimate-guide-to-face-recognition/

9. Facial Recognitions and Privacy Rights

Limitations of facial recognition

Limitations	Description
Accuracy	Facial recognition does not have the same degree of accuracy as fingerprint or DNA identification,
Facial features	The frequency that facial features occur in the general population is unknown
Image quality	Quality of images taken may affect the comparisons.
Stored images	The size of the cohort of database images for comparison my affect the facial recognition accuracy.
Changes of facial features over time.	Changes in an individual facial's face over time, could result in false positive or negative matches.

9. Facial Recognitions and Privacy Rights

9.2 Privacy Rights

- The expanding use of biometric facial recognition raises a number of ethical concerns.
- The concerns relate especially to the **potential conflicts between security**, on the one hand, **and individual privacy** on the other.

9. Facial Recognitions and Privacy Rights

9.2 Privacy Rights

Privacy is a right that people have in relation to other persons, the state and organisations with respect to:

(a) **The possession of information (including facial images) about themselves by other persons and by organisations,** e.g. personal information and images stored in biometric databases, **or;**

(b) **The observation/perceiving of themselves** – including of their movements and relationships – by other persons.

9. Facial Recognitions and Privacy Rights

9.2 Privacy Rights

Biometric facial recognition is obviously implicated in both informational and observational concerns.

10. Fingerprint and Applied Ethics

10.1 Biometric identification

- The measurement of physical aspects of the human body.
- This can include patterns of the skin or blood vessel networks under the skin; patterns in the genetic code; facial appearance, such as the distance between features such as the eyes, nose or mouth.

10. Fingerprint and Applied Ethics

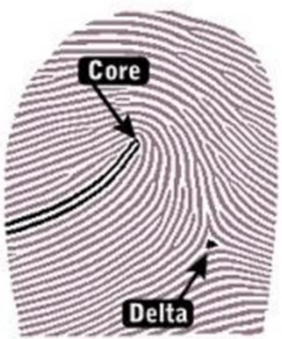
10.1 Biometric identification (cont.)

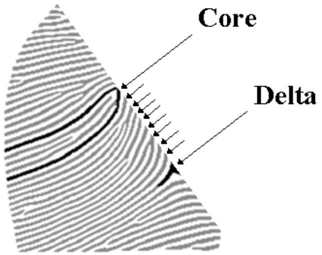
- Reasons for using biometric identification:
  - Biometrics is unique between individuals.
  - Unchanging over time.
  - Capable of being digitalised through an algorithm
  - It can be converted to a format that can be integrated with automated database storage and searching.
  - Biometric identification can be contrasted with other methods of identification, such as keys, identification cards and passwords.

10. Fingerprint and Applied Ethics

10.1.1 Fingerprint identification

- The technique of fingerprint identification, in both analogue and digital forms, is based on differences within the standard patterns of the ridges.
- These can be classified into a series of arches, loops and whorls.
- The centre of a pattern is referred to as the core, and points of deviation are referred to as the delta.

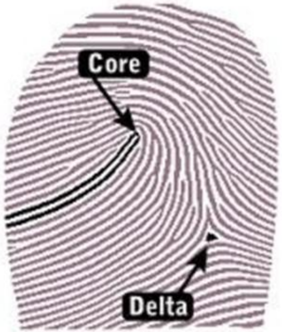


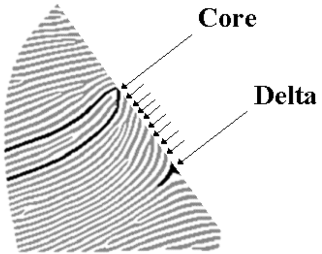


10. Fingerprint and Applied Ethics

10.1.1 Fingerprint identification

- The fingerprint identification has been widely used outside law enforcement, with the first major development being the integration of biometric fingerprint identification (along with facial recognition) into passports and border control systems.





11. Smartphone Applications

- Fingerprint and facial recognition biometrics are now widely used to identify and grant access to a smartphone.
- Due to the high level of security these biometrics provide, possession of a smartphone registered to a specific person has become a proxy for the identity or location of that person.
- For example, a smartphone can now be used as a tap and pay device, in the same way as a credit or debit card has been used in the past, it can be used to record the presence of a person at a location, using quick response (QR) code scanning, and provides access to social media and other online accounts.

11. Smartphone Applications

- The accuracy and security of biometric identification technologies have enabled smartphones to become an extension of the physical self for identification purposes.

12. Social Media

- Biometric fingerprint and facial recognition, in regulating access to smartphones, simultaneously provide access to social media accounts, and are therefore key indicators of a person's identity in online environments.
- Facial recognition is widely used to identify and link individuals within social media platforms, such as Facebook's tagging feature.

12. Social Media

- Social media does not include all online websites, but involves a degree of **interaction between participants**, and collaboration in a non-hierarchical way. It enables users to post self-generated content, such as text and photos; allows users to create profiles and engage with others by posting comments or ‘likes’; and, enables users to network with others that hold similar interests or opinions.

12. Social Media

- Technology companies such as Google and Facebook have become powerful due to the vast amount of data they holding detailing the internet activity of their billions of users.
- How information available on the internet is presented to users also has a significant **capacity to influence social views and trends**.
- In contrast with traditional mediums, there is a relative lack of central control over content that can facilitate mistruths to be perpetuated.

12. Social Media

- It has recently been proposed in a number of countries around the world, that social media users must provide evidence of their identity, such as a copy of a passport or drivers licence in order to obtain, or maintain a social media account.
- The objective of this approach is to address the issue of people using anonymous accounts to harass and abuse online: described as ‘technology facilitated abuse’, or commit other crimes.

12. Social Media

- In an anonymous online environment, harsh comments can be widely observed on public social media websites.
- Online harassment may target individuals or groups on the basis of race, ethnicity, religion or sexual orientation, and is widespread, with recent survey data indicating that one in three have experienced some form of online harassment impacting their health, safety and productivity.

12. Social Media

- There are other issues arising from social media that may also be mitigated with the introduction of identity verification measures such as dissemination of misleading or inaccurate information or theories, commonly referred to as ‘fake news’, that can rely on automated dissemination using botnets: such as misinformation in relation to the COVID-19 pandemic.
- Other issues including: Use social media advertisements specifically targeting their personality and social views to influence their vote during the presidential election campaign.

12. Social Media

- Social media can significantly impact the lives of individuals and the nature of society.
- Social media is too powerful now to be anonymous' and that just as identification and registration is required to drive a car or own a firearm, so it should also be required to operate a social media account.

12. Social Media

- To date, laws requiring compulsory identity verification for social media account holders have not been introduced. They could plausibly deter online harassment and abuse, hate speech and disinformation and enable it to be better investigated and prosecuted.

12. Social Media

- Potential issues with the identity verification for social media:
  - data security, if identity documents, such as copies of passports and drivers licences, were provided to multinational technology companies such as Google and Facebook, which already have a great deal of personal data about users online and real world (e.g. location metadata) behaviour, they would be a target for organised crime groups, and would increase the level of risk associated with the already detailed and sensitive information that social media companies hold about individuals.

13. Major laws on Privacy

Some federal laws governing privacy matters have been created.

For example,

- **Gramm-Leach-Bliley Act** protects personal financial information
- **Health Insurance Portability and Accountability Act (HIPAA)** protects medical records
- **Family Educational Rights and Privacy Act (FERPA)** restricts disclosure of educational records.

Most of the information collected by private organizations is not covered by existing laws. However, as more and more individuals become concerned about controlling who has the right to personal information and how that information is used, companies and lawmakers will respond.

Question?

"Ethics is knowing the difference between what you have a right to do and what is right to do." - Potter Stewart

Disclaimer:

The slide content has been sourced from the following reference books and Internet resources. Refer to the following list for more details and further reading:

1. Awari, G.K., & Warjukar, S.V. (2022). *Ethics in Information Technology: A Practical Guide*. CRC Press, Taylor & Francis Group.
2. Smith, M., & Miller, S. (2021). *Biometric Identification, Law and Ethics*. Springer.
3. O'Leary, T., O'Leary, L., & O'Leary, D. (2024). Computing Essentials 2025: 2024 Release ISE. McGrawHill LLC. https://tarc.vitalsource.com/reader/books/9781264543182
4. Hasselfeld, B. (2020). *Benefits of Telemedicine.* https://www.hopkinsmedicine.org/health/treatment-tests-and-therapies/benefits-of-telemedicine
5. Dan C Marinescu (2022). *Cloud Computing: Theory and Practice.* Morgan Kaufmann.
6. https://www.forensicsciencesimplified.org/

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